

● EASY

● GEOMETRY AND TRIGONOMETRY

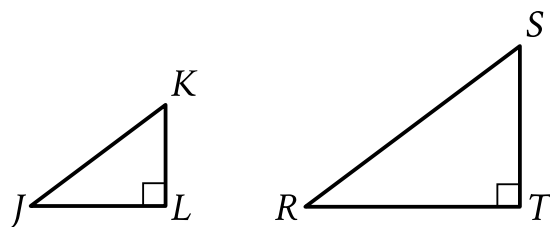
Area and volume

03

9 questions · ~14 min

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Right triangle upper J upper K upper L is labeled as follows:
 - Angle upper L is a right angle.
- Right triangle upper R upper S upper T is labeled as follows:
 - Angle upper T is a right angle.
- A note indicates the figures are not drawn to scale.

In the figure shown, triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . The length of $\bar{JK}$ is 15, and the perimeter of triangle JKL is 36. The length of $\bar{RS}$ is 135. What is the perimeter of triangle RST ?

- A. 324
- B. 540
- C. 2,916
- D. 8,100

Q2

GEOMETRY AND TRIGONOMETRY

Triangle R has an area of 80 square centimeters cm^2 . Square S has side lengths of 4 cm. What is the total area of triangle R and square S, in cm^2 ?

- A. 42
- B. 44
- C. 84
- D. 96

Q3

GEOMETRY AND TRIGONOMETRY

A rectangle has a length of 13 and a width of 6. What is the perimeter of the rectangle?

- A. 12
- B. 26
- C. 38
- D. 52

Q4

GEOMETRY AND TRIGONOMETRY

A rectangle has an area of 63 square meters and a length of 9 meters. What is the width, in meters, of the rectangle?

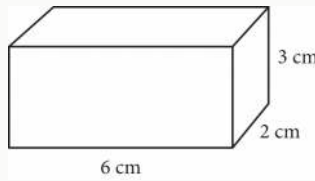
- A. 7
- B. 54
- C. 81
- D. 567

Q5

GEOMETRY AND TRIGONOMETRY

What is the area of a rectangle with a length of 17 centimeters cm and a width of 7 cm?

- A. 24 cm^2
- B. 48 cm^2
- C. 119 cm^2
- D. 576 cm^2



The figure shows the lengths, in centimeters (cm), of the edges of a right rectangular prism. The volume V of a right rectangular prism is ℓwh , where ℓ is the length of the prism, w is the width of the prism, and h is the height of the prism. What is the volume, in cubic centimeters, of the prism?

- A. 36
- B. 24
- C. 12
- D. 11

Q7

GRID-IN

GEOMETRY AND TRIGONOMETRY

A triangle has a base length of 40 centimeters and a height of 90 centimeters. What is the area, in square centimeters, of the triangle?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GEOMETRY AND TRIGONOMETRY

The area of a rectangle is 630 square inches. The length of the rectangle is 70 inches. What is the width, in inches, of this rectangle?

- A. 9
- B. 70
- C. 315
- D. 560

Q9

GEOMETRY AND TRIGONOMETRY

The table gives the perimeters of similar triangles TUV and XYZ , where TU corresponds to XY . The length of TU is 6.

	Perimeter
Triangle TUV	50
Triangle XYZ	150

What is the length of XY ?

- A. 2
- B. 6
- C. 18
- D. 56